



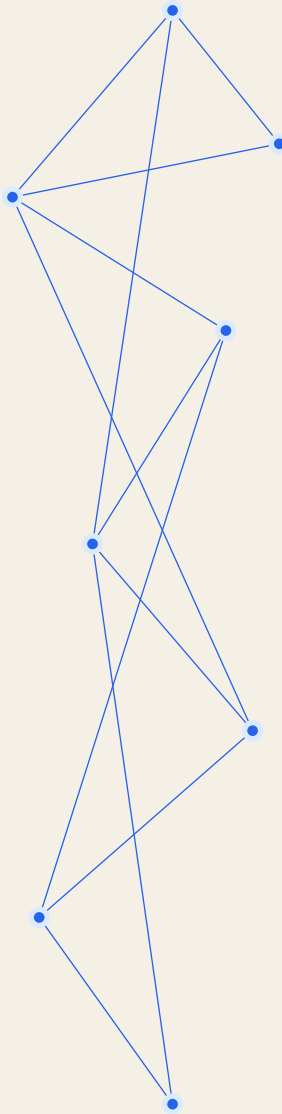
# NightGuard AI

## AI-Powered Insider Threat Investigation Platform

Detecting suspicious employee behaviour and surfacing explainable, evidence-backed alerts before data breaches occur.

EVIL AI

Logapriya K · Kiruthiga S · Monika S · Kaviya T  
*Hackathon Submission*



# Team

LK

**Logapriya K**  
Data Engineering

KS

**Kiruthiga S**  
Model Development

MS

**Monika S**  
Backend Integration

KT

**Kaviya T**  
Frontend Dashboard

## Hackathon Focus

Rapid prototype of an end-to-end investigator for insider threat detection — from log ingestion to explainable AI investigation reports.

# ■ Problem Statement

Organisations face rising risk from insider threats: employees downloading sensitive data, using unauthorised USB devices, accessing confidential files, or working at unusual hours.

**Traditional monitoring tools produce noisy alerts without investigation support — wasting analyst time and delaying response to real threats.**

**USB**

Unauthorised device usage

**Bulk**

Large sensitive downloads

**Off-hrs**

Unusual access timing

# The Need

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**S**

## **Ingest Activity Logs**

Continuously collect employee activity logs across endpoints, file servers, and USB events.

**C**

## **Detect Anomalies**

Identify behaviour that deviates from established historical baselines.

**R**

## **Automated Investigation**

Perform root-cause investigation automatically using AI, without manual triage.

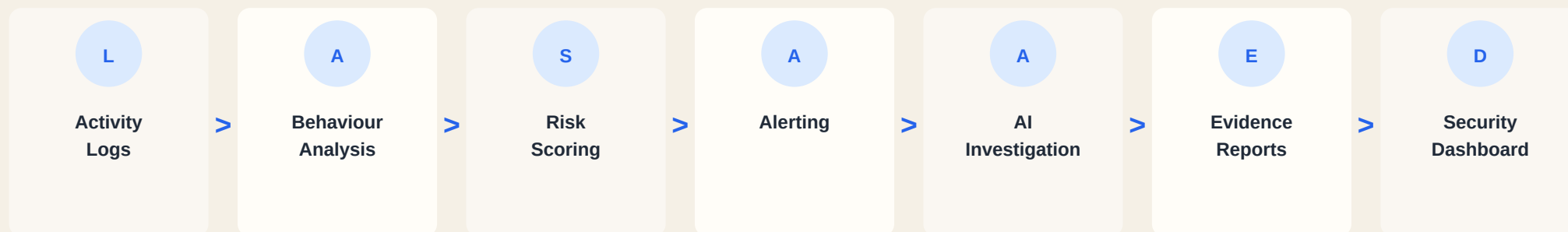
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## **Actionable Reports**

Produce evidence packages and reports for rapid remediation by SOC teams.

# Solution Overview

NightGuard AI is an end-to-end insider threat investigation platform that monitors activity logs, detects abnormal behaviour, scores risk, generates explainable AI investigation reports, and produces evidence packages — all surfaced in a security dashboard.



*Designed for explainability and an evidence-first workflow, so SOC teams can prioritise and act with confidence.*

# Architecture

End-to-end pipeline architecture covering data ingestion through to dashboard visualisation:



# Methodology

1

Collect activity logs from endpoints, file servers, and USB events

2

Compare download volumes and access patterns to historical baselines

3

Detect signals: large downloads, USB usage, sensitive-file access, out-of-hours activity

4

Compute risk scores and classify threat level (Low to Critical)

5

Run AI Investigation Engine to produce explainable assessments and evidence

6

Visualise incidents and timelines on the dashboard

# ■ Implemented Features



## Large Download Detection

Flag abnormal bulk transfers relative to baseline



## USB Usage Monitoring

Track device attachments and data exfiltration attempts



## Sensitive File Access

Detect attempts to open or copy protected documents



## Out-of-Hours Activity

Identify sessions outside normal working patterns



## Risk Scoring Engine

Aggregate signals into calibrated severity tiers



## AI Investigation & Evidence

Auto-generated, explainable investigation reports and evidence packages



# Dashboard — Threat Overview

## NightGuard AI — Security Dashboard

2

CRITICAL

0

HIGH

0

MEDIUM

3

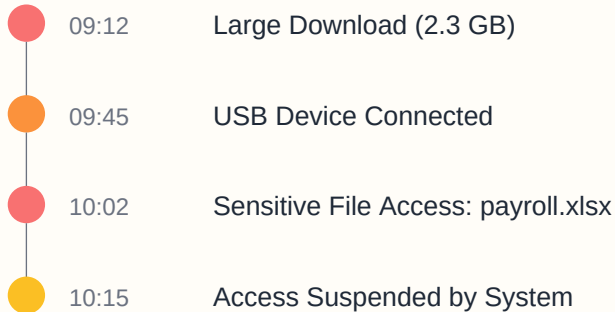
LOW

| Employee | Risk Score | Threat Level | AI Confidence | Actions                     |
|----------|------------|--------------|---------------|-----------------------------|
| Bob      | 180        | CRITICAL     | 100%          | <a href="#">View Report</a> |
| David    | 100        | CRITICAL     | 100%          | <a href="#">View Report</a> |
| Alice    | 10         | LOW          | 95%           | <a href="#">View Report</a> |
| Charlie  | 10         | LOW          | 92%           | <a href="#">View Report</a> |
| Emma     | 10         | LOW          | 90%           | <a href="#">View Report</a> |

The dashboard surfaces severity counts at a glance and lists every monitored employee with their risk score, AI confidence, and threat level.

# Dashboard — Investigation Detail

## ■ Threat Timeline — Bob



## ■ AI Investigation Summary

- Bob:** HIGH RISK — Large download, USB, sensitive file access. Possible data exfiltration.
- David:** HIGH RISK — Unusual access patterns after hours detected.
- Alice:** Normal activity. No anomalies detected.
- Charlie:** Normal activity. No anomalies detected.
- Emma:** Normal activity. No anomalies detected.

## Threat Timeline

Reconstructs the exact sequence of suspicious actions per employee — large download, USB connection, sensitive file access, and resulting access suspension — with timestamps.

## AI Investigation Summary

Each employee receives a plain-language assessment — flagging Bob and David as potential insider threats while confirming Alice, Charlie, and Emma appear normal.

# Dashboard — Risk Ranking & Alerts

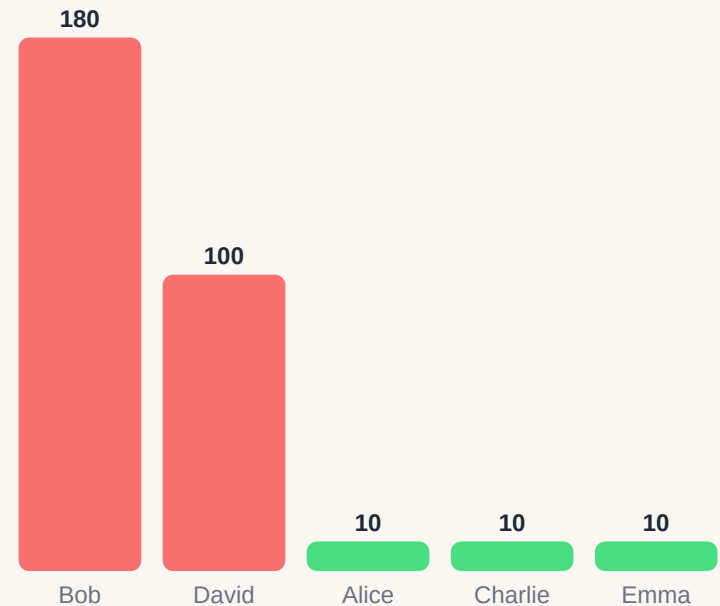
## Employee Risk Ranking

|    |         |     |          |
|----|---------|-----|----------|
| #1 | Bob     | 180 | CRITICAL |
| #2 | David   | 100 | CRITICAL |
| #3 | Alice   | 10  | LOW      |
| #4 | Charlie | 10  | LOW      |
| #5 | Emma    | 10  | LOW      |

## Recent Alerts

|          |                                           |       |
|----------|-------------------------------------------|-------|
| CRITICAL | Bob — Large download 2.3 GB detected      | 09:12 |
| CRITICAL | Bob — USB device connected during session | 09:45 |
| CRITICAL | David — Off-hours access: 02:34 AM        | 02:34 |
| LOW      | Alice — Normal login activity             | 08:55 |

## Risk Score Comparison



### Bob (Score: 180)

Ranked #1 critical threat. Bob and David hold CRITICAL status with 100% AI confidence.

## Results (Prototype)

2

Critical Risk  
Employees

0

High Risk  
Employees

0

Medium Risk  
Employees

3

Low Risk  
Employees

**Detected critical-risk employees: Bob and David.**

Outputs produced automatically: alert files, AI investigation reports, an evidence package, and a security summary.

The prototype successfully correlated anomalous actions and produced forensic-ready reports.

# Future Enhancements & Differentiators

1



## Real-time Monitoring

Stream logs and evaluate risk with sub-minute latency

2



## ML Anomaly Models

Unsupervised learning to surface subtle deviations

3



## Enterprise Integrations

SIEM, IAM, endpoint agents, cloud connectors

4



## Automated Notifications

Email/SMS and SOAR playbooks for faster response

5

## Predictive Intelligence

Forecast risk trends and proactively harden controls

NightGuard AI emphasises explainability and an evidence-first workflow so SOC teams can prioritise and act with confidence.



# Thank You

**NightGuard AI — AI-Powered Insider Threat Investigation Platform**

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